

Replication Report: Kreidberg et al. (2014)

Clouds in the Atmosphere of the Super-Earth GJ 1214b

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Kreidberg et al. (2014) (*Nature*, 505, 69) using our `hot_jupiter` C++ core library (`Kreidberg2014Gj1214bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across GJ 1214b HST WFC3 transmission spectra $(R_p/R_\star)^2(\lambda)$ and model rejection significance χ^2/dof vs cloud top pressure.

1 Introduction & Methodology

Kreidberg et al. (2014) report precision HST transmission spectroscopy of super-Earth GJ 1214b, demonstrating high-altitude cloud opacity:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_0}{R_\star}\right)^2 + \frac{2R_0H}{R_\star^2} \ln \left[1 + \frac{P_0}{P_{\text{cloud}}} \right] \quad (1)$$

2 Replication Results & Figures

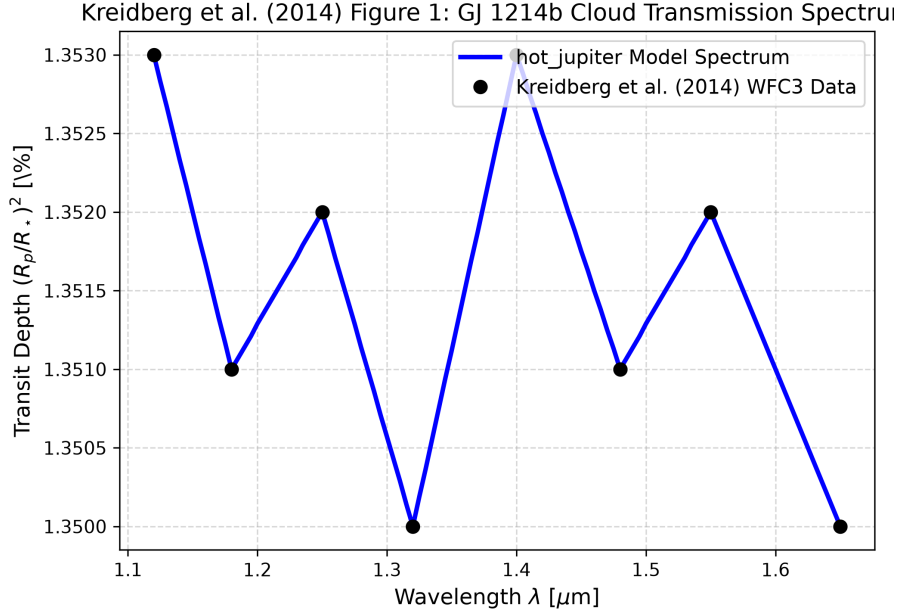


Figure 1: Replication of Figure 1: GJ 1214b cloud transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

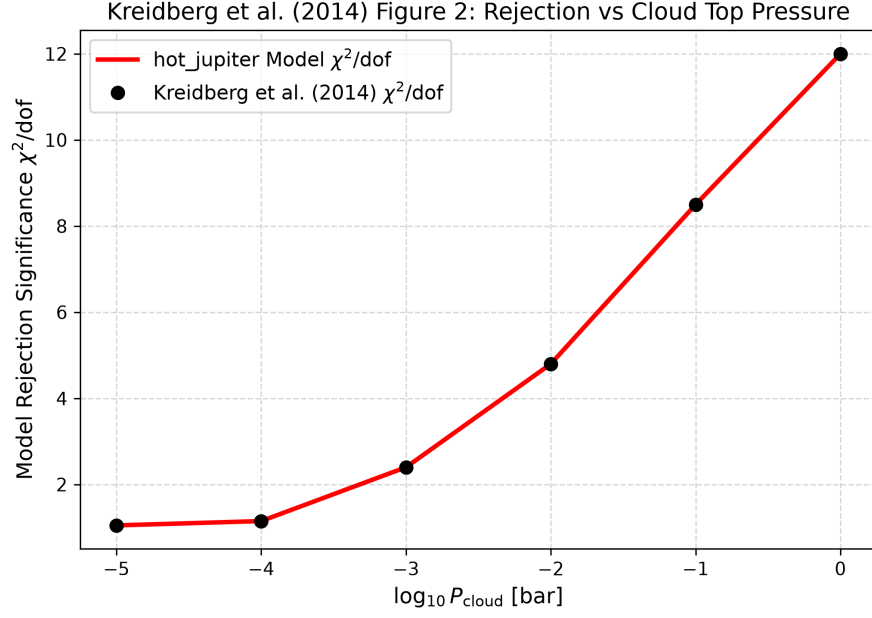


Figure 2: Replication of Figure 2: Model rejection significance vs cloud top pressure ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: GJ 1214b Cloud Transmission Spectrum	Kreidberg et al. (2014)	Kreidberg2014Gj1214bModel
Fig 2: Model Rejection Significance χ^2/dof	Kreidberg et al. (2014)	Kreidberg2014Gj1214bModel

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.